Chapter: 3
Basic of Machine Learning

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What is Learning?

“To gain **knowledge** or **understanding** of, or **skill** in **by study**, **instruction** or **experience**"

- Learning a set of new facts.
- Learning HOW to do something .
- Improving ability of something already learned.

What is Machine Learning?

- Machine Learning is the study of methods for programming computers to learn.
- Building machines that automatically learn from experience.
- Machine learning usually refers to the changes in systems that perform tasks associated with artificial intelligence AI Such tasks involve recognition, diagnosis, planning, robot control, prediction, etc.

What is Machine Learning?

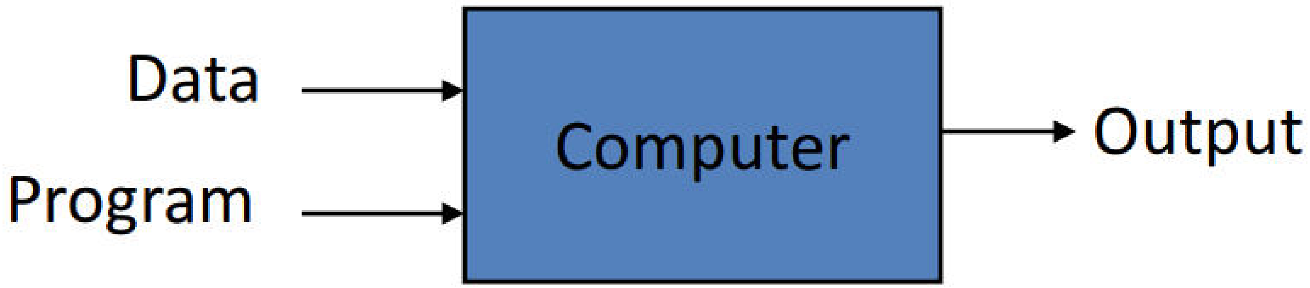
- Definition by Tom Mitchell (1998): Machine Learning is the study of algorithms that
  - improve their performance P
  - at some task T
  - with experience E.

A well-defined learning task is given by **<P,T,E>.**

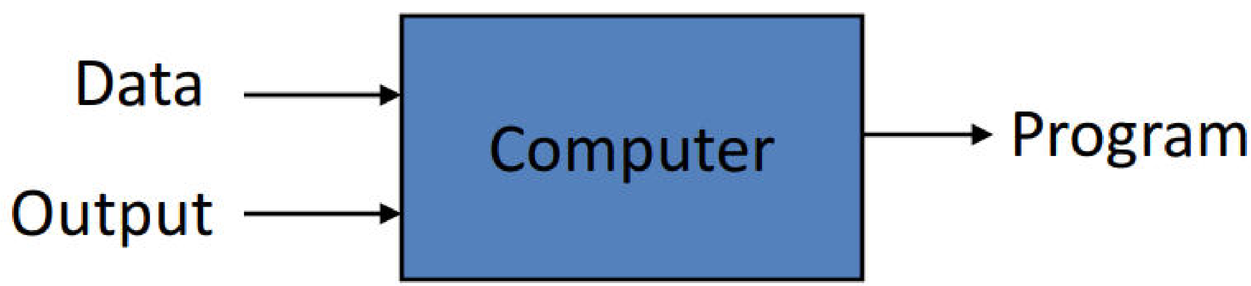
- "Learning is any process by which a system improves performance from experience." - Herbert Simon

What is Machine Learning?

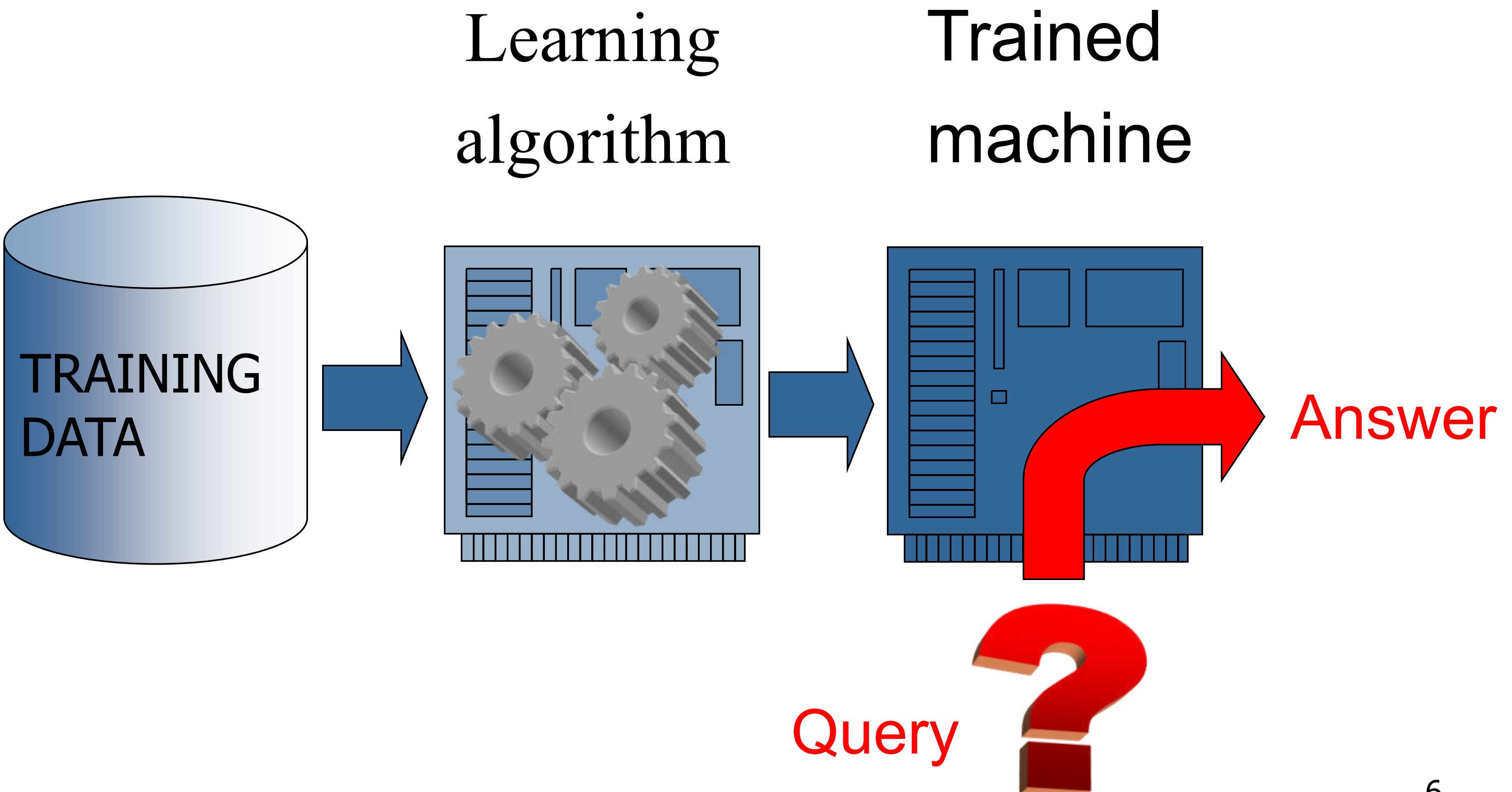
- Traditional Programming



- Machine Learning



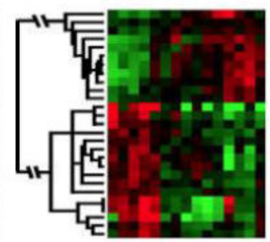
What is Machine Learning?



When Do We Use Machine Learning?

- ML is used when:
- Human expertise does not exist (navigating on Mars)
- Humans can’t explain their expertise (speech recognition)
- Models must be customized (personalized medicine)
- Models are based on huge amounts of data (genomics)

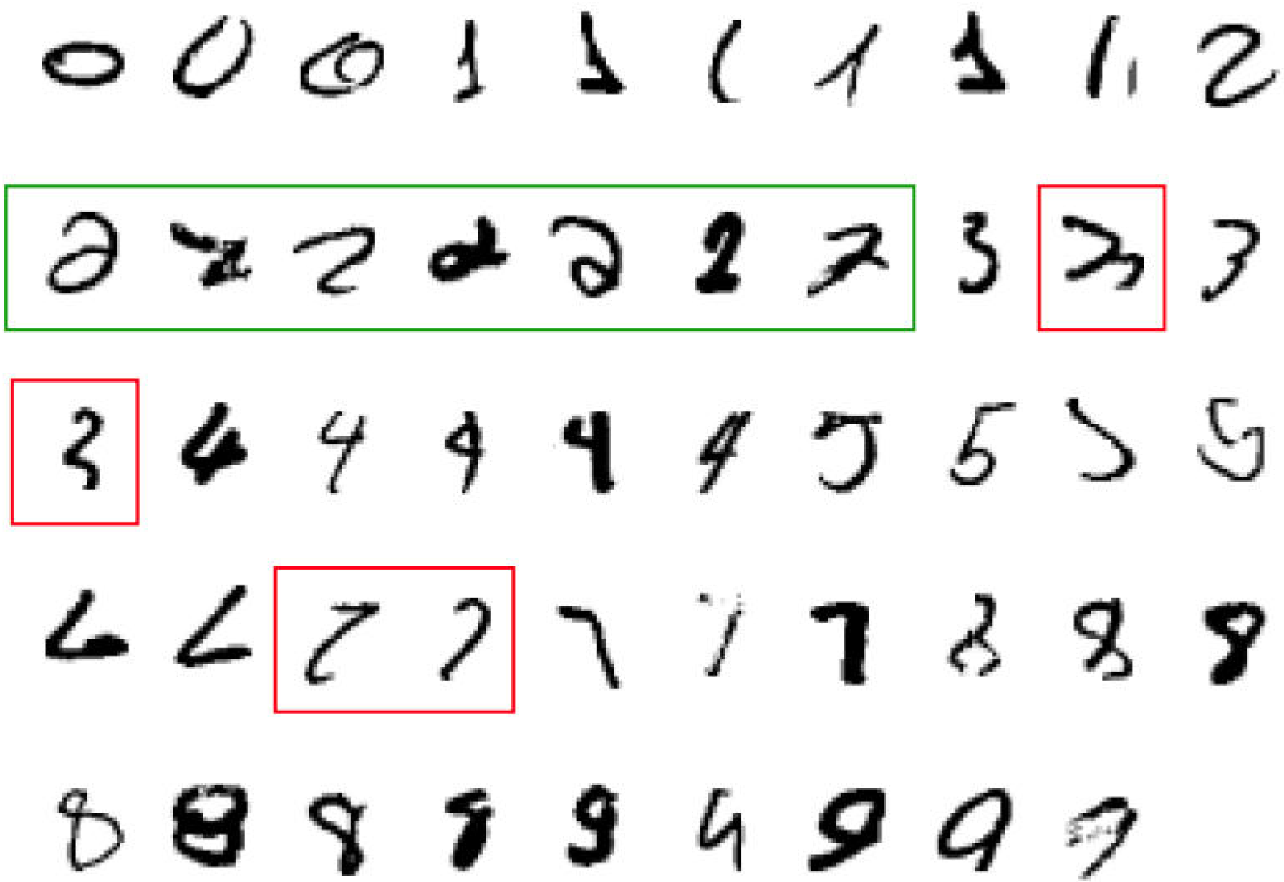


- Learning isn’t always useful:
- There is no need to “learn” to calculate payroll

Examples

- A classic example of a task that requires machine learning: It is very hard to say what makes a 2



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Some more examples of tasks that are best solved by using a learning algorithm

- Recognizing patterns:
  - Facial identities or facial expressions
  - Handwritten or spoken words
  - Medical images
- Generating patterns:
  - Generating images or motion sequences
- Recognizing anomalies:
  - Unusual credit card transactions
  - Unusual patterns of sensor readings in a nuclear power plant
- Prediction:
  - Future stock prices or currency exchange rates

Types of machine Learning

1) Supervised Learning.

2) Unsupervised Learning .

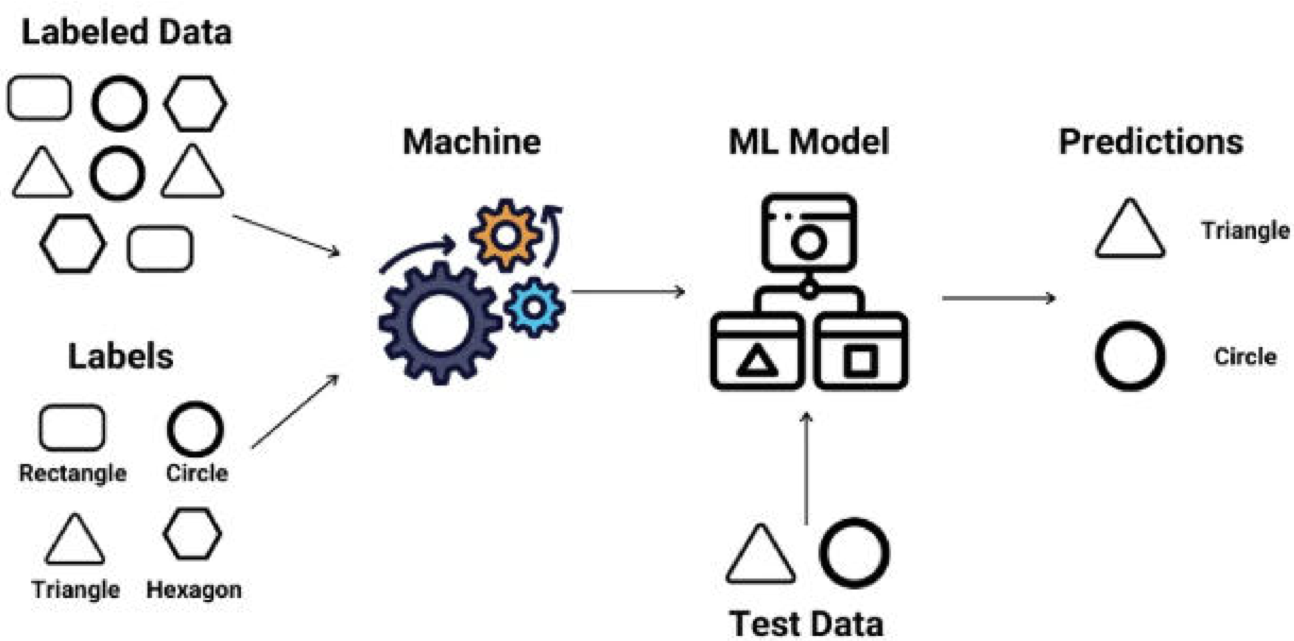
3) Semi-Supervised (reinforcement).

- Supervised learning as the name indicates has the presence of a **supervisor** as a teacher.
- Supervised learning is when we teach or train the machine using data that is well-labelled. Which means some data is already tagged with the correct answer.
- After that, the machine is provided with a new set of examples (data) so that the supervised learning algorithm analyses the training data (set of training examples) and produces a correct outcome from labeled data.

Supervised Learning

1) Supervised learning

Given: training data + desired outputs (labels)



Types of Supervised Learning

- Supervised learning is classified into two categories of algorithms:
- **Regression**: A regression problem is when the output variable is a real value, such as "dollars" or "weight".
- **Classification**: A classification problem is when the output variable is a category, such as "Yes" or "No" , "disease" or "no disease".

1. Regression

- Regression is a type of supervised learning that is used to predict continuous values, such as house prices, stock prices, or customer churn. Regression algorithms learn a function that maps from the input features to the output value.

Some common regression algorithms include:

- Linear Regression
- Polynomial Regression
- Lasso Regression
- Ridge Regression

2. Classification

- Classification is a type of supervised learning that is used to predict categorical values, such as whether a customer will churn or not, whether an email is spam or not, or whether a medical image shows a tumor or not. Classification algorithms learn a function that maps from the input features to a probability distribution over the output classes.

Some of the most common classification algorithms include:

- Logistic Regression
- Support Vector Machines
- Decision Trees
- Random Forests
- Naive Baye

Advantages of Supervised learning

- By using labeled examples, supervised learning builds models that draw on prior experiences to produce reliable outputs for new, unseen data.

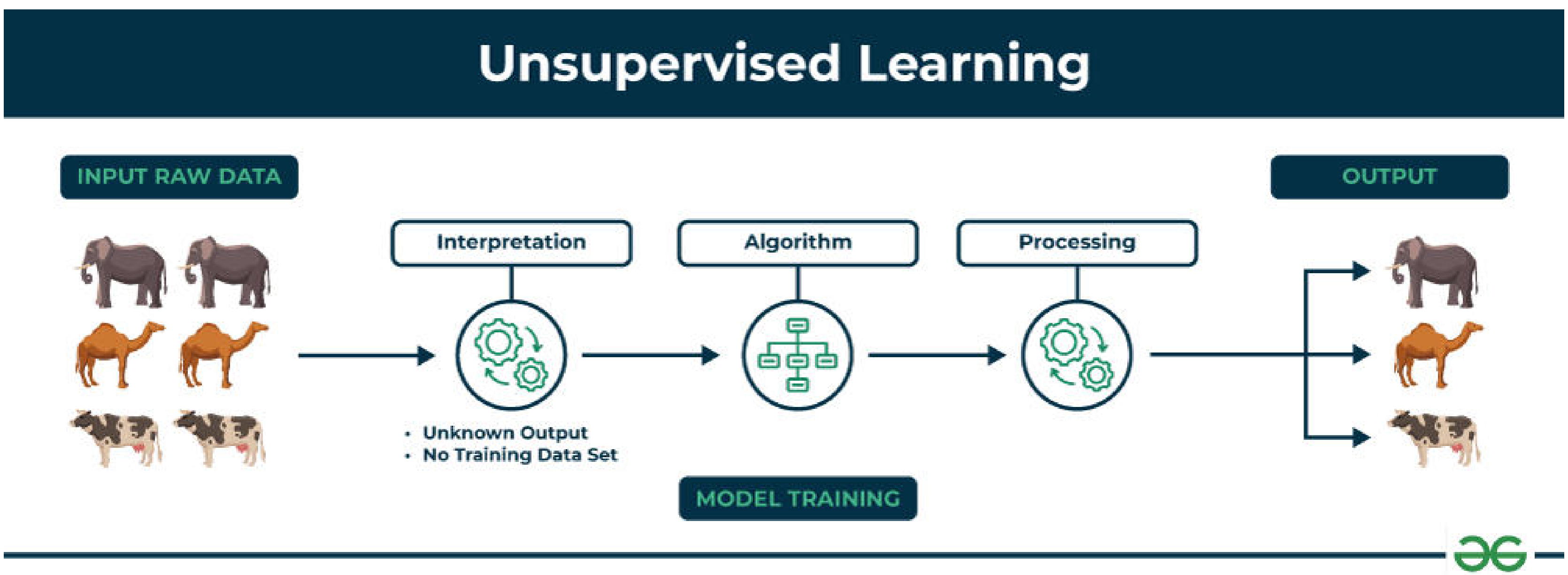
- It can be used to categorize data into predefined labels (classification) or to predict continuous outcomes (regression), providing flexibility for various tasks.

Unsupervised Learning

- Unsupervised learning is a type of machine learning that works with data that has **no labels or categories.**
- The main goal is to **find patterns** and **relationships** in the data without any guidance.
- **No outputs** are used (unlike supervised learning and reinforcement learning)
- Learner is provided only unlabeled data.
- **No feedback** is provided from the environment

Cont.

- Unlike supervised learning, there is **no teacher** or training involved. The machine must uncover hidden structures in the data on its own.
- For example, unsupervised learning can analyze animal data and group the animals by their traits and behavior.
- These groups could correspond to different species, making it possible to organize the animals without pre-existing labels.



Unsupervised Learning

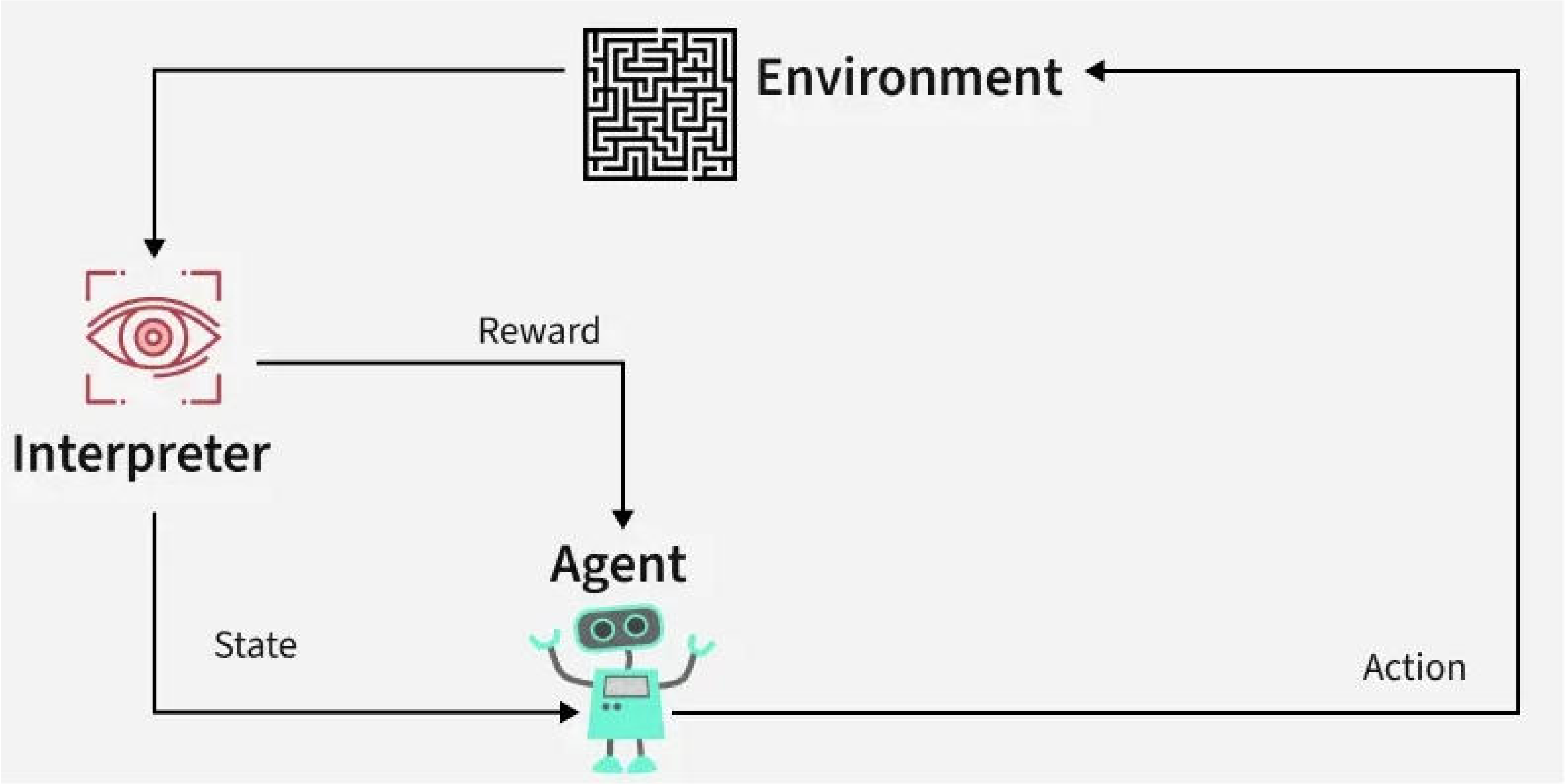
- **Advantage**
- Most of the **laws of science** were developed through unsupervised learning.

- **Disadvantage**
- The identification of the features itself is a complex problem in many situations.

Reinforcement

- **Reinforcement Learning (RL)** is a branch of machine learning that focuses on how agents can **learn** to make decisions through trial and error to maximize cumulative rewards.
- RL allows machines to learn by interacting with an environment and receiving **feedback** based on their actions.
- This feedback comes in the form of **rewards or penalties**.
- Reinforcement Learning revolves around the idea that an agent (the learner or decision-maker) interacts with an environment to achieve a goal.
- The agent performs actions and receives feedback to optimize its decision-making over time.

Reinforcement

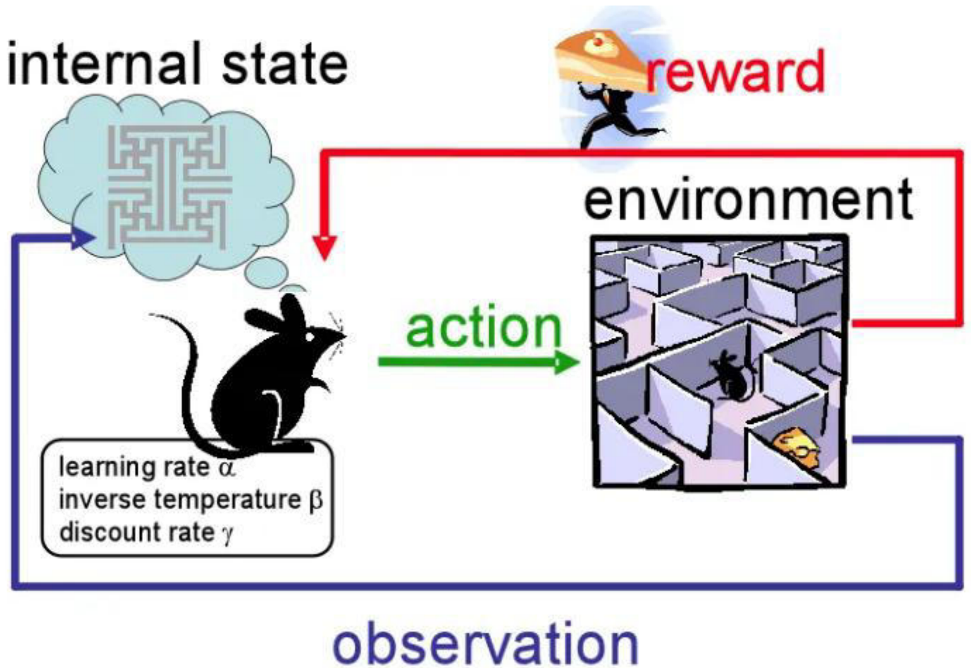


Reinforcement

- **Agent**: The decision-maker that performs actions.
- **Environment**: The world or system in which the agent operates.
- **State**: The situation or condition the agent is currently in.
- **Action**: The possible moves or decisions the agent can make.
- **Reward**: The feedback or result from the environment based on the agent's action.

Cont.

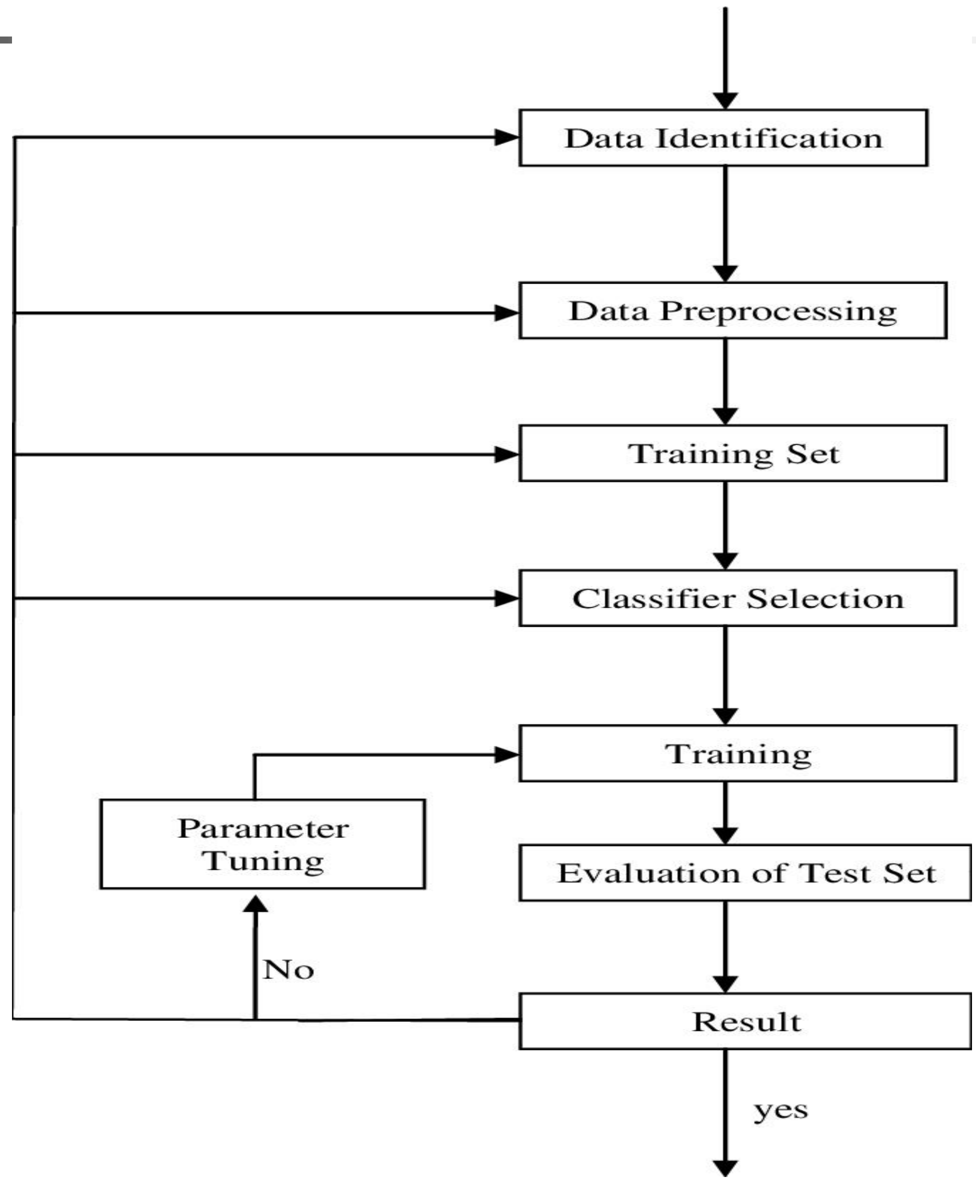
- The RL process involves an agent performing actions in an environment, **receiving rewards or penalties** based on those actions, and **adjusting its behavior accordingly.**
- This loop helps the agent improve its decision-making over time to maximize the **cumulative reward**.



Application of Reinforcement Learning

- Robotics
- Game Playing
- Industrial Control
- Personalized Training Systems

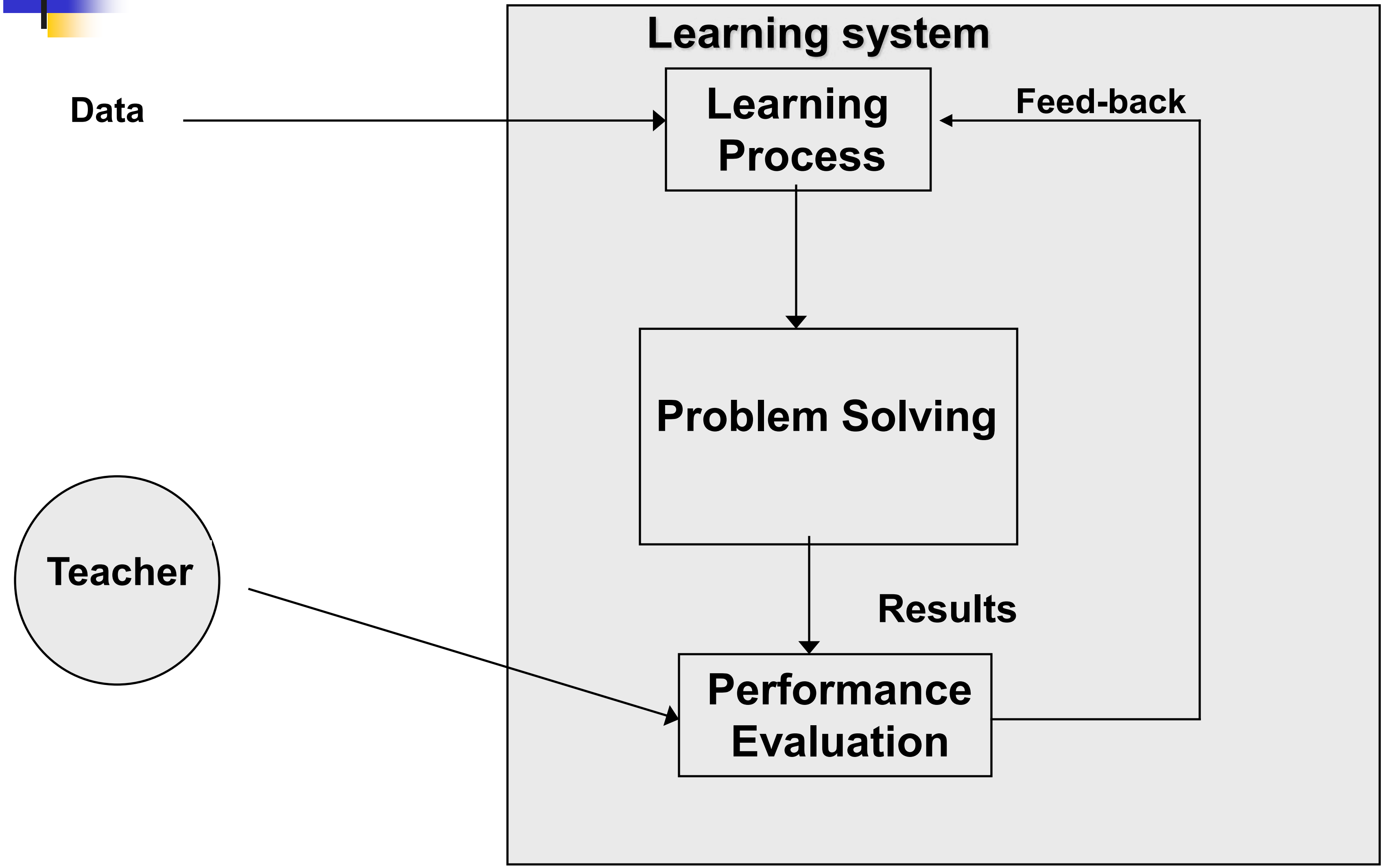
Cont.


Data Identification
Data Preprocessing
Training Set
Classifier Selection
Training
Parameter Tuning
Evaluation of Test Set
No
Result
yes

Steps in machine learning

1) Data collection.
2) Representation.
3) Modeling.
4) Estimation.
5) Validation.
6) Apply learned model to new “test” data

General structure of a learning system



Learning by Decision Tree

- A decision tree receives a set of attributes (or properties) of the objects as inputs and yields a binary decision of true or false values as output.
- Decision trees, thus, generally represent Boolean functions. Besides a range of {0,1}other non-binary ranges of outputs are also allowed.
- However, for the sake of simplicity, we presume the restriction to Boolean outputs. Each node in a decision tree represents 'a test of some attribute of the instance, and each branch descending from that node corresponds to one of the possible values for this attribute'

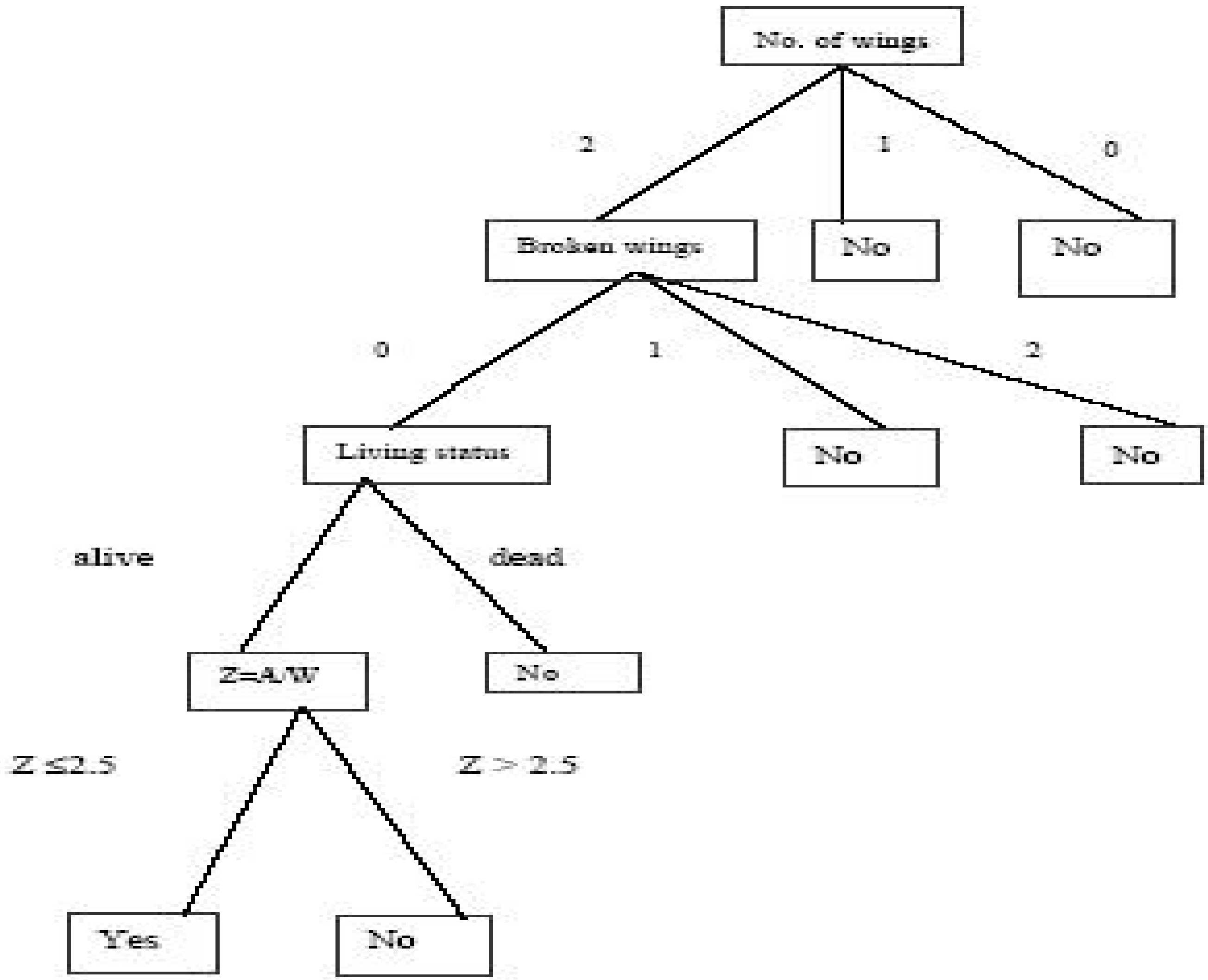
Learning by Decision Tree

To illustrate the contribution of a decision tree, we consider a set of instances, some of which result in a true value for the decision. Those instances are called **positive instances. On the other hand, when the resulting decision** is false, we call the instance '**a negative instance'. We now consider** the learning problem of a bird's flying. Suppose a child sees different instances of birds as tabulated below.

Learning by Decision Tree

Instances	No. of wings	Broken wings if any	Living status	Wing area/ weight of bird	Fly
1.	2	0	alive	2.5	True
2.	2	1	alive	2.5	False
3.	2	2	alive	2.6	False
4.	2	0	alive	3.0	True
5.	2	0	dead	3.2	False
6.	0	0	alive	0	False
7.	1	0	alive	0	False
8.	2	0	alive	3.4	True
9.	2	0	alive	2.0	False

Decision Tree example



Decision Tree example

It is seen from the above table that Fly = true if (no. of wings=2)$\wedge$ (broken wings=0)$\wedge$(living status-alive)$\wedge$((wing area / weight) of the bird $\geq$2.5) is true.

Thus we can write:

Fly = (no. of wings = 2) $\wedge$ (broken wings = 0) $\wedge$ (living status = alive) $\wedge$ (wing area / weight (A/W) $\geq$ 2.5)

Advantages of ML

1) Solving vision problems through statistical inference.

2) Intelligence from the common sense AI.

3) Reducing the constraints over time achieving complete autonomy.

Disadvantages of ML

1) Application specific algorithms.

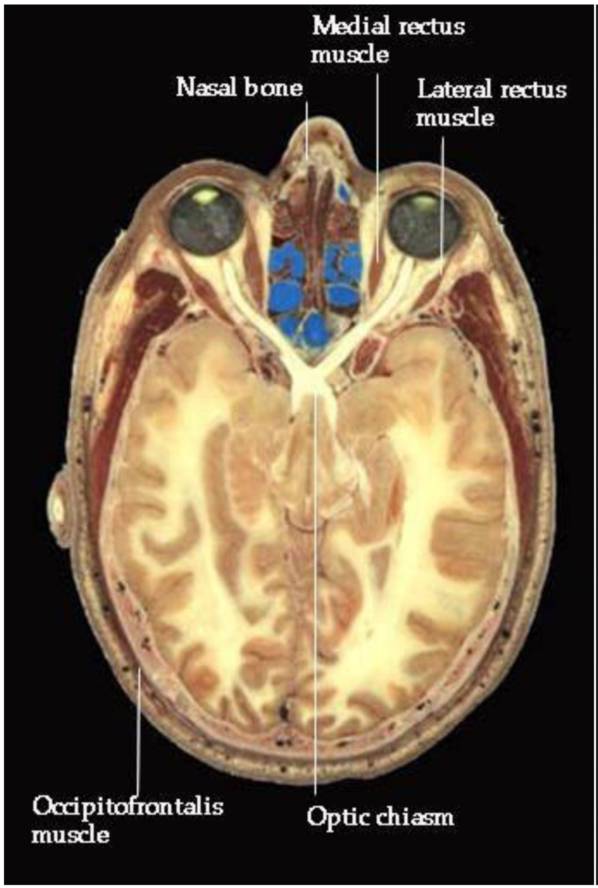
2) Real world problems have too many variables and sensors might be too noisy.

3) Computational complexity.

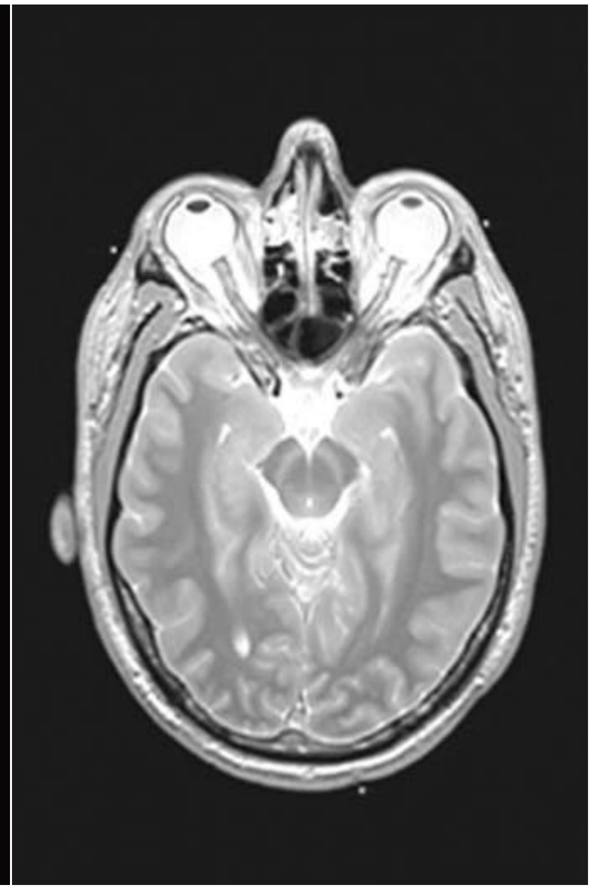
Applications of Machine Learning

Drug discovery

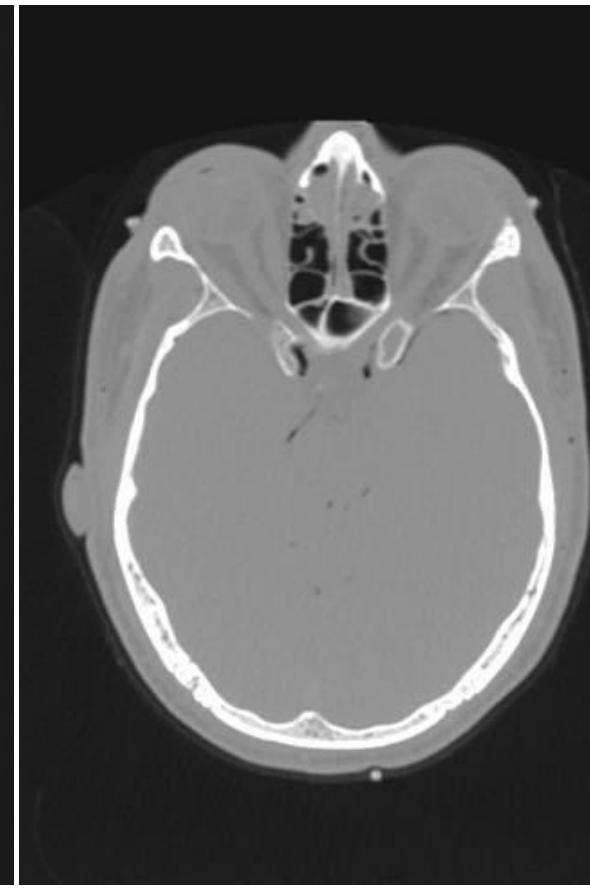
Medical diagnosis



Photo

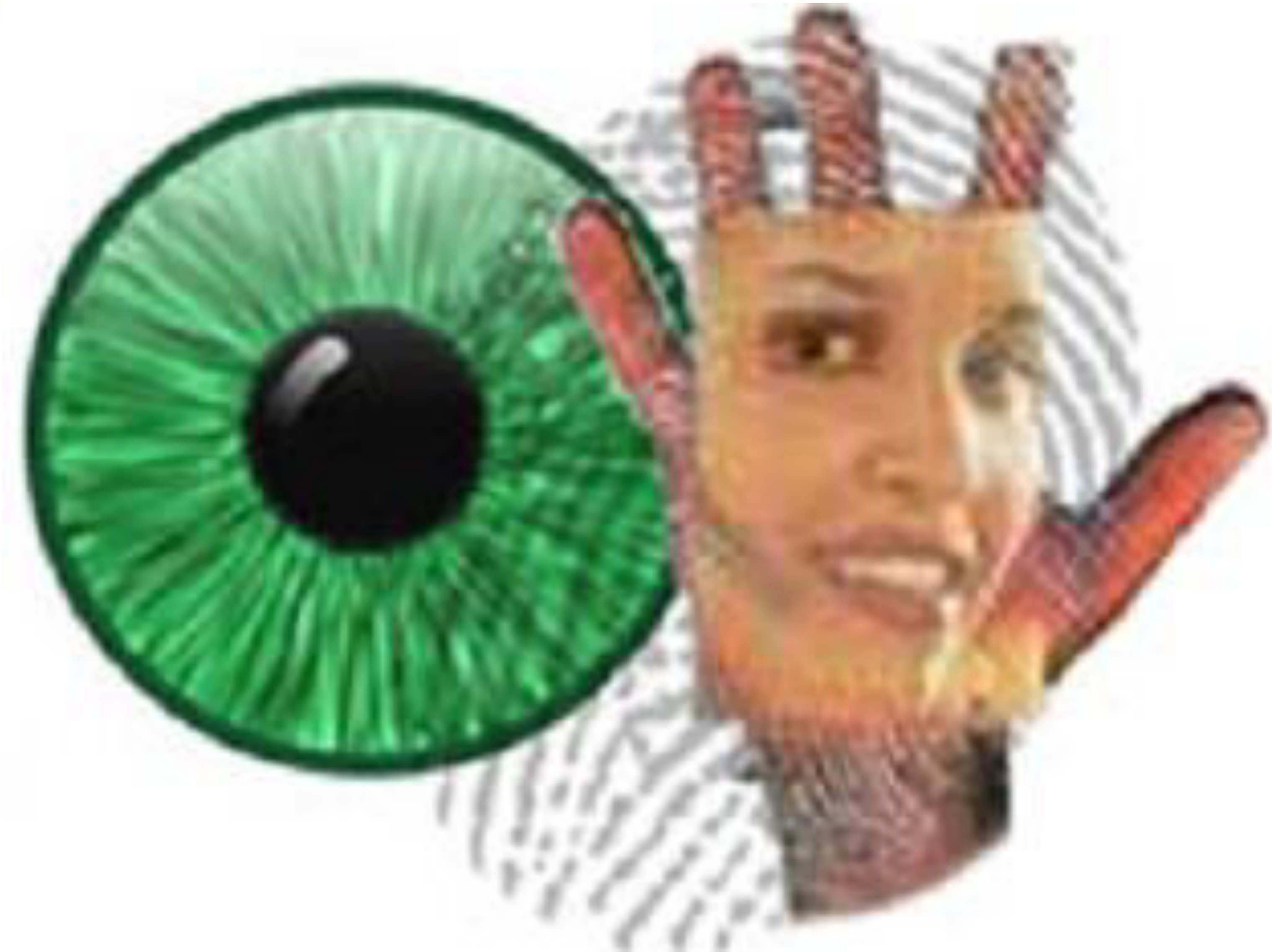


MRI



CT

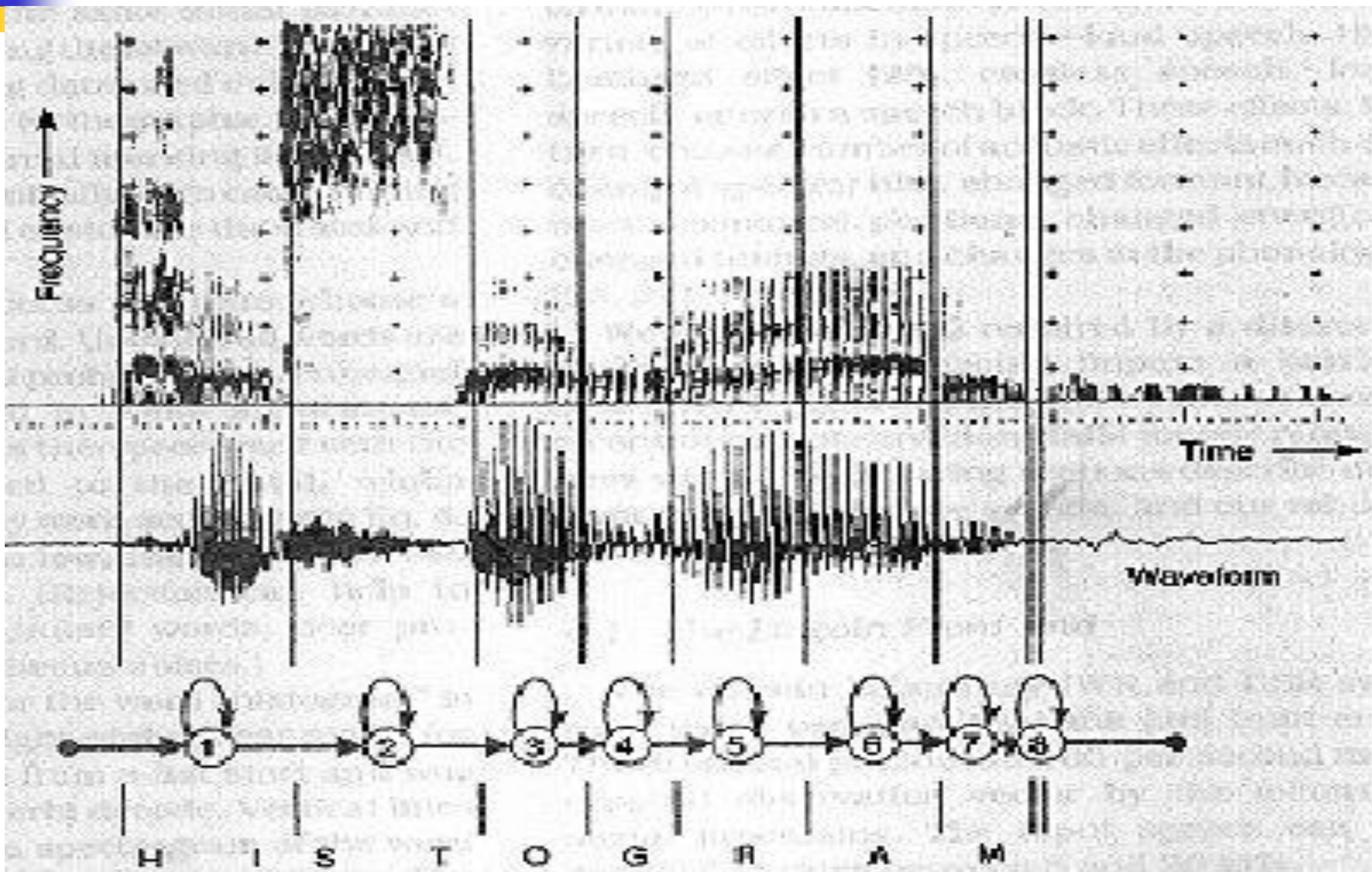
Iris verification



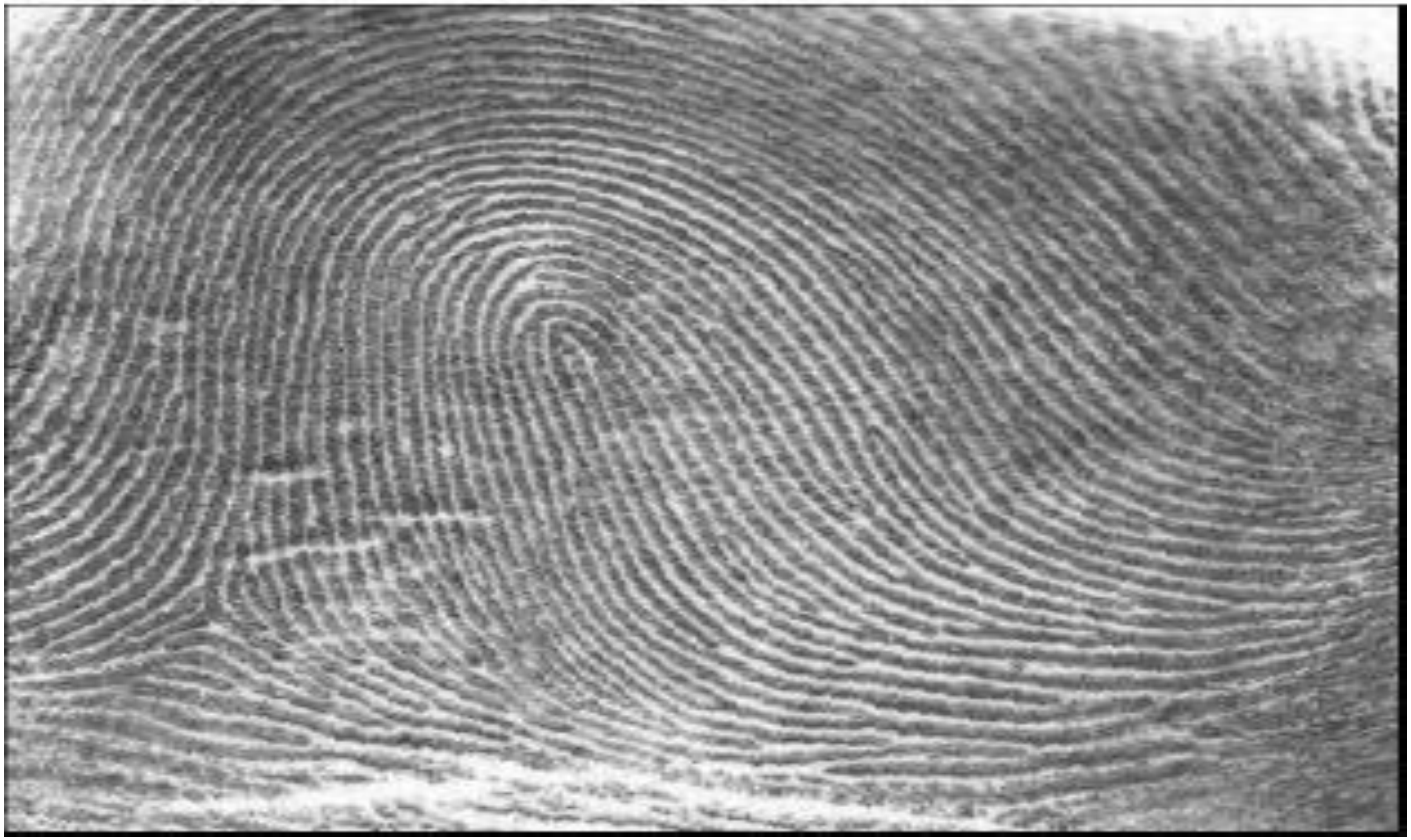
Hand-written digits

Radar Imaging

Speech Recognition

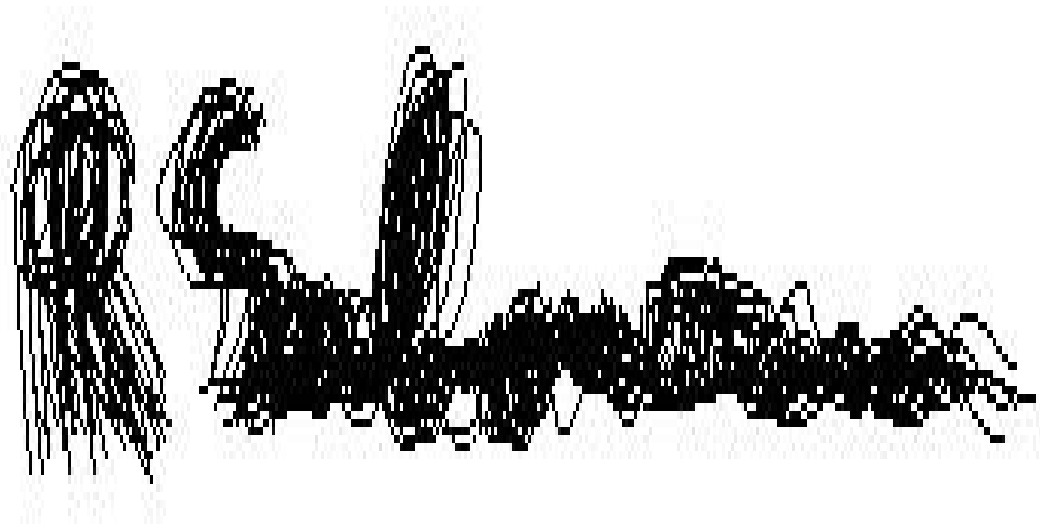


Finger print



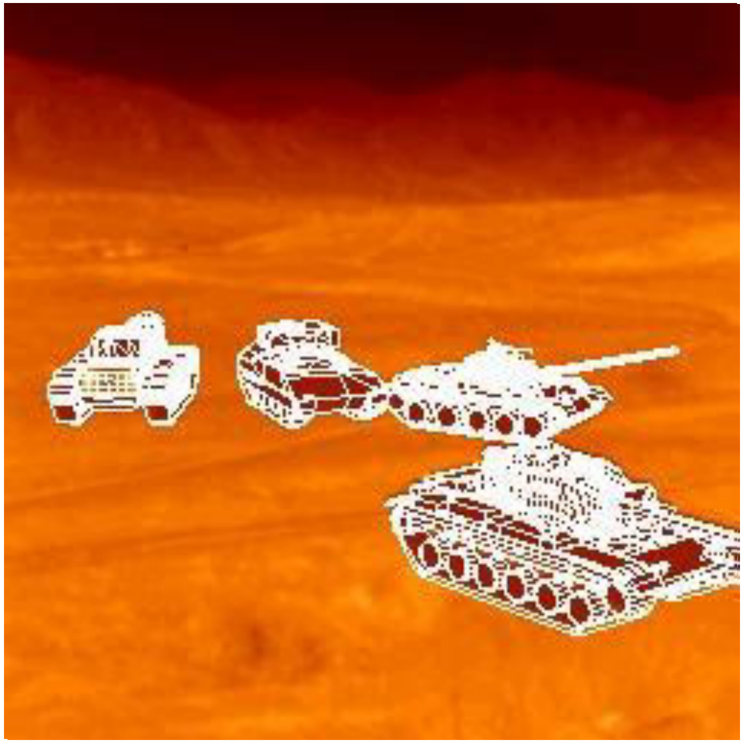
fingerprint image

Signature Verification



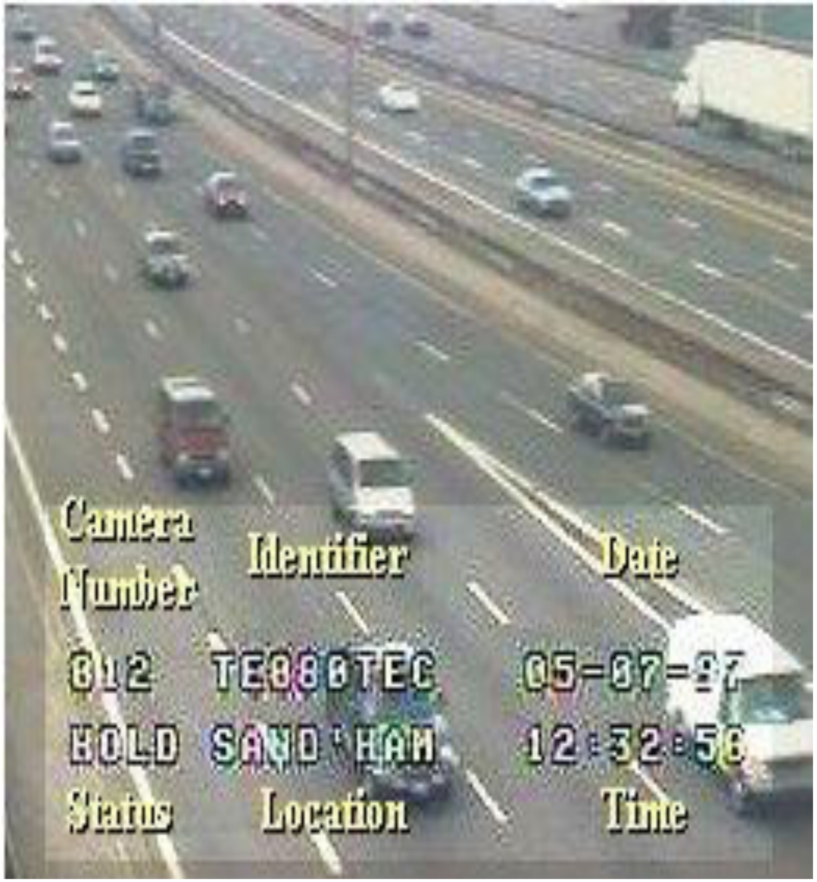
Face Recognition

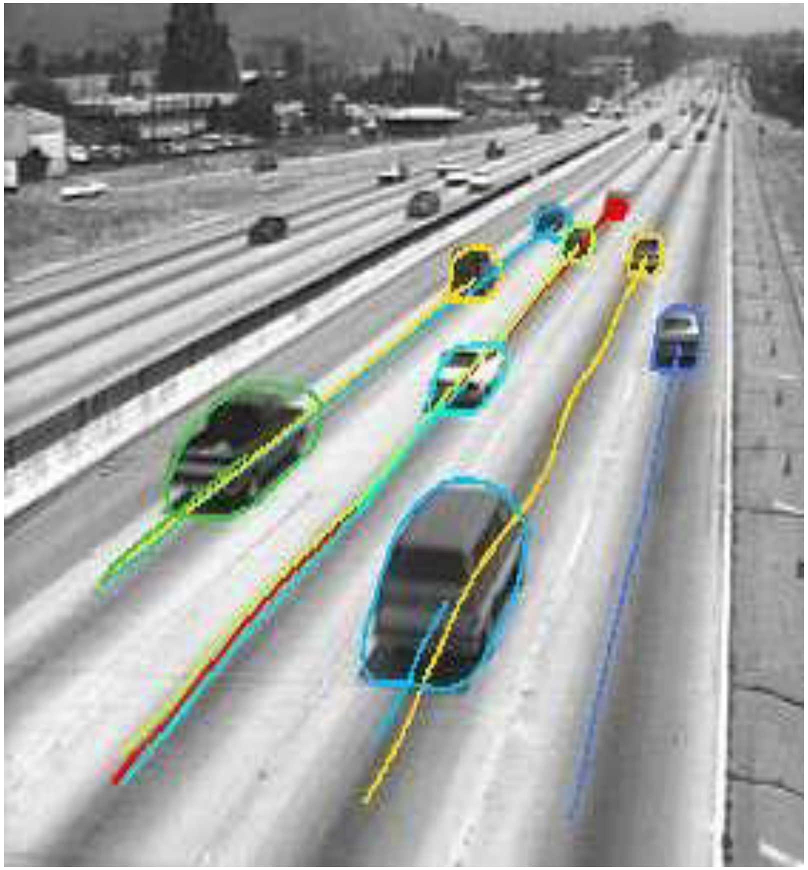
Target Recognition



Robotics vision

Traffic Monitoring





Thank you